

量子與統計

2021.10.06

一. 算符的構成

概念: 算符作用於波函數使其變成另一個函數的運算符號。力學量 F 的算符為 $\hat{F}$

基本假定及構成: 在量子力學中任一可觀測力學量 F 可以用厄米(Hermite)算符 $\hat{F}$ 表示, 並作用於態的波函數。

坐標算符: $\hat{r} = r$

動量算符: $\hat{p} = -i\hbar \nabla$

力學量 $F = f(r, p) \Rightarrow \hat{F} = f(\hat{r}, \hat{p}) = f(r, -i\hbar \nabla)$

角動量算符: $\hat{L} = \hat{r} \times \hat{p} = \hat{r} \times (-i\hbar \nabla)$

非相對論動能算符 $\hat{T} = \frac{\hat{p}^2}{2m} = -\frac{\hbar^2}{2m} \nabla^2 = -\frac{\hbar^2}{2m} \Delta$

勢能算符: $\hat{V} = V$

系統總能量算符:

能量算符 $\hat{E} = \hbar \omega \frac{\partial}{\partial t}$

哈密頓算符 $\hat{H} = \hat{T} + \hat{V}$

在量子理論中, 時間作為連續變化的參量, 所以不存在“時間算符”

二. 算符運算規則 (Ψ, Φ 任意波函數, $C_1, C_2 \in \mathbb{C}$)

1. 線性: $A(C_1\Phi + C_2\Psi) = C_1A\Phi + C_2A\Psi$

* 量子力學中的算符不都是線性算符 (例如複共軛), 但刻量可觀察測量的算符都是線性的

2. 算符相等: $A\Psi = B\Psi \Rightarrow A=B$

3. 單位算符: $I\Psi = \Psi$

4. 算符之和: $(A+B)\Psi = A\Psi + B\Psi$

交換律: $\hat{A} + \hat{B} = \hat{B} + \hat{A}$

結合律: $(\hat{A} + \hat{B}) + \hat{C} = \hat{A} + (\hat{B} + \hat{C})$

5. 算符之積: $(\hat{A}\hat{B})\Psi = \hat{A}(\hat{B}\Psi)$, $\hat{A}\hat{B} \neq \hat{B}\hat{A}$

例子: $\hat{x}\hat{p}_x \neq \hat{p}_x\hat{x}$

$(\hat{x}\hat{p}_x)\Psi = \hat{x}(\hat{p}_x\Psi) = -i\hbar x \frac{d\Psi}{dx} = -i\hbar x \frac{d\Psi}{dx}$

$(\hat{p}_x\hat{x})\Psi = \hat{p}_x(x\Psi) = -i\hbar \frac{d}{dx}(x\Psi) = -i\hbar (x \frac{d\Psi}{dx} + \Psi)$

$(\hat{x}\hat{p}_x - \hat{p}_x\hat{x})\Psi = -i\hbar \Psi \neq 0$

6. 冪運算: $\hat{A}^n = \underbrace{\hat{A}\hat{A} \dots \hat{A}}_{n \text{ 個 } \hat{A}}$

7. 逆算符: 若 $A\Psi = \Phi$ 能唯一地求出 Ψ , 則定義 $\hat{A}^{-1}$, $\Psi = \hat{A}^{-1}\Phi$

滿足矩陣乘法

8. 算符的共轭

$\hat{A}$ 的共轭算符 $\hat{A}^\dagger$

$$\hat{p}_x = (-i\hbar \frac{d}{dx})^\dagger = i\hbar \frac{d}{dx} = -\hat{p}_x$$

9. 算符的厄米共轭和厄米算符

$\hat{A}$ 的 Hermitian conjugate: $\hat{A}^\dagger$

$$\int \psi^\dagger \hat{A} \phi dx = \int [\hat{A} \psi]^\dagger \phi dx$$

$$(\hat{A}^\dagger)^\dagger = \hat{A}$$

$$(\hat{A} + \hat{B})^\dagger = \hat{A}^\dagger + \hat{B}^\dagger$$

$$(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger$$

若任意 ψ, ϕ , 若算符 $\hat{F}$ 满足

$$\int \psi^\dagger (\hat{F}\phi) dx = \int (\hat{F}\psi)^\dagger \phi dx, \text{ 则 } \hat{F} \text{ 为 Hermite 算符}$$

若 $\hat{F}^\dagger = \hat{F}$, 则 $\hat{F}$ 是自共轭算符

~~位置~~ $x, p_x, \hat{L}, \hat{H} (V \in \mathbb{R}), \hat{P}$ (空间反射) 都是 Hermite 算符

$$\begin{aligned} \int \psi^\dagger (\hat{A}^\dagger)^\dagger \phi dx &= \int [\hat{A}^\dagger \psi]^\dagger \phi dx = [\int \phi^\dagger \hat{A}^\dagger \psi dx]^* = [\int [\hat{A} \phi]^\dagger \psi dx]^* \\ &= \int \psi^\dagger \hat{A} \phi dx \end{aligned}$$

$$\Rightarrow \int \psi^\dagger [(\hat{A}^\dagger)^\dagger - \hat{A}] \phi dx = 0 \Leftrightarrow (\hat{A}^\dagger)^\dagger = \hat{A}$$

$$\begin{aligned} \int \psi^\dagger (\hat{A}\hat{B})^\dagger \phi dx &= \int [\hat{A}\hat{B} \psi]^\dagger \phi dx = \int [\hat{A}(\hat{B}\psi)]^\dagger \phi dx \\ &= \int (\hat{B}\psi)^\dagger \hat{A}^\dagger \phi dx = \int \psi^\dagger (\hat{B}^\dagger \hat{A}^\dagger) \phi dx \Leftrightarrow \int \psi^\dagger [(\hat{A}\hat{B})^\dagger - \hat{B}^\dagger \hat{A}^\dagger] \phi dx = 0 \\ &\Rightarrow (\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger \end{aligned}$$

Hermite 算符的本征值.

(1) 都是实数:

$$\hat{F}\psi = \lambda\psi$$

$$\int \phi^\dagger (\hat{F}\psi) dx = \int (\hat{F}\phi)^\dagger \psi dx$$

$$\text{取 } \phi = \psi$$

$$\text{则 } \lambda \int \psi^\dagger \psi dx = \lambda^* \int \psi^\dagger \psi dx$$

$$\Rightarrow \lambda = \lambda^*$$

~~定理 (2): 体系的任何~~

(2) 体系的任何状态, 其厄米算符的平均值均为实数.

$$(\psi, \psi) = \int \psi^\dagger \psi dx$$

$$\bar{A} = \int \psi^\dagger \hat{A} \psi dx = \int (\hat{A} \psi)^\dagger \psi dx = [\int \psi^\dagger (\hat{A} \psi) dx] = \bar{A}^*$$

↓

逆定理: 在任何狀態下平均值均為實的算符必為 Hermite 算符

$$\bar{A} = \bar{A}^* \quad (\psi, \hat{A}\psi) = (\psi, \hat{A}\psi)^* = (\hat{A}\psi, \psi)$$

證明在 Page 16-17

Hermite 算符之和 $\in$ Hermite 算符

Hermite 算符之積 不一定為 Hermite 算符

(9) Unitary (幺正) 算符 $\hat{A}^\dagger = \hat{A}^{-1}$

例 $\hat{p}$

$$\int_{-\infty}^{+\infty} \psi^*(x) \hat{p} \phi(x) dx = \int_{-\infty}^{+\infty} \psi^*(x) (-i\hbar \frac{d}{dx}) \phi(x) dx = \int_{-\infty}^{+\infty} \phi(x) \psi^*(x) dx = \int_{-\infty}^{+\infty} (\hat{p} \psi)^* \phi dx$$

$$\Rightarrow \hat{p} = \hat{p}^\dagger = \hat{p}^{-1}$$

自共轭厄米正算符

(10) 算符的函數

若 $F(x)$ 可被 Taylor 展開, 則 $F(\hat{A})$ 也可被 Taylor 展開

若 $\hat{A}\psi_n = a_n\psi_n$

$$F(\hat{A})\psi_n = \sum_{m=0}^{\infty} \frac{F^{(m)}(0)}{m!} \hat{A}^m \psi_n = \left(\sum_{m=0}^{\infty} \frac{F^{(m)}(0)}{m!} a_n^m \right) \psi_n = F(a_n) \psi_n$$

$$F^{(n,m)}(x,y) = \frac{\partial^n}{\partial x^n} \frac{\partial^m}{\partial y^m} F(x,y)$$

$$F(x,y) = \sum_{n,m \in \mathbb{N}^2} \frac{F^{(n,m)}(0,0)}{n!m!} x^n y^m$$

對易: $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$, 若 $[\hat{A}, \hat{B}] = 0$, 則 $\hat{A}, \hat{B}$ 對易

量子統計

2021. 10. 08

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A} \quad \{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}$$

$$[\hat{x}, \hat{y}] = [\hat{x}, \hat{p}_y] = [\hat{p}_x, \hat{p}_y] = 0$$

$$[\hat{x}, \hat{p}_x] = [\hat{y}, \hat{p}_y] = [\hat{z}, \hat{p}_z] = i\hbar$$

$$\hat{L} = \hat{r} \times \hat{p} = \hat{r} \times (i\hbar \nabla) = -i\hbar \left(\begin{vmatrix} \hat{y} & \hat{z} \\ \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{vmatrix}, \begin{vmatrix} \hat{z} & \hat{x} \\ \frac{\partial}{\partial z} & \frac{\partial}{\partial x} \end{vmatrix}, \begin{vmatrix} \hat{x} & \hat{y} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \end{vmatrix} \right)$$

$$= \hbar (y\hat{p}_z - z\hat{p}_y, z\hat{p}_x - x\hat{p}_z, x\hat{p}_y - y\hat{p}_x)$$

$$[\hat{L}_x, \hat{L}_y] = [y\hat{p}_z - z\hat{p}_y, z\hat{p}_x - x\hat{p}_z]$$

$$= [y\hat{p}_z - z\hat{p}_y, z\hat{p}_x] + -[y\hat{p}_z - z\hat{p}_y, x\hat{p}_z]$$

$$= [y\hat{p}_z, z\hat{p}_x] + [z\hat{p}_y, x\hat{p}_z]$$

$$= y[\hat{p}_z, z\hat{p}_x] + x[z\hat{p}_y, \hat{p}_z]$$

$$= (x\hat{p}_y - y\hat{p}_x)[z, \hat{p}_z] = i\hbar(x\hat{p}_y - y\hat{p}_x)$$

$$= i\hbar \hat{L}_z$$

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$$

$$[\hat{L}_x, \hat{L}^2] = [\hat{L}_y^2, \hat{L}^2] = [\hat{L}_z^2, \hat{L}^2] = 0$$

$$[\hat{L}_x, \hat{x}_y] = \epsilon_{xyx} i\hbar x_x \quad \begin{cases} \epsilon_{xyz} = -\epsilon_{zyx} = -\epsilon_{xzy} \\ \epsilon_{123} = 1 \end{cases}$$

$$[\hat{L}_x, \hat{p}_y] = \epsilon_{xyx} i\hbar p_x$$

共同本征函数

完全系 $\{\phi_n\}$: $\forall \psi \in$ 這個系統, $\psi \in \mathbb{C}$, $\psi = \sum_n C_n \phi_n$

若 $\hat{F}, \hat{G}$ 有一組共同本征函数, 且 $\{\phi_n\}$ 是完全系. 則 $\hat{F}, \hat{G}$ 對易

$$\hat{F}\phi_n = \lambda_n \phi_n, \quad \hat{G}\phi_n = \mu_n \phi_n$$

$$(\hat{F}\hat{G} - \hat{G}\hat{F})\phi_n = \hat{F}(\mu_n \phi_n) - \hat{G}(\lambda_n \phi_n) = \lambda_n \mu_n \phi_n - \mu_n \lambda_n \phi_n = 0$$

$$\Rightarrow (\hat{F}\hat{G} - \hat{G}\hat{F})\psi = \sum_n C_n (\hat{F}\hat{G} - \hat{G}\hat{F})\phi_n = 0$$

若 $\hat{F}, \hat{G}$ 對易, 則 $\hat{F}, \hat{G}$ 有共同本征函数

證: 非簡並, $\hat{F}\phi_n = \lambda_n \phi_n$

$$\hat{F}\hat{G}\phi_n = \hat{G}\hat{F}\phi_n = \lambda_n \hat{G}\phi_n$$

$$\Rightarrow \hat{G}\phi_n = \mu_n \phi_n$$

用 Schmit 正交化, 令 $\{\phi_n\}$ 正交归一

$$\hat{F}\phi_n = \lambda_n \phi_n, \quad \psi_n = \sum_{\alpha} C_{\alpha} \phi_n$$

~~$\hat{F}\phi_n$~~

$$\hat{F}\hat{G}\phi_n = \hat{G}\hat{F}\phi_n = \lambda_n \hat{G}\phi_n$$

$$\hat{G}\phi_n = \sum_{\alpha} C_{\alpha} \phi_{n\alpha}, \quad C_{\alpha} = (\phi_{n\alpha}, \hat{G}\phi_n)$$

$$\psi(x, y, z) = (2\pi\hbar)^{-\frac{3}{2}} \exp\left(\frac{i}{\hbar} \vec{p}_0 \cdot \vec{r}\right)$$

$$[\hat{L}^2, \hat{L}_z] = 0, \text{ 同时本征函数 } Y_{lm}(\theta, \varphi) (?)$$

$$\text{氢原子 } [\hat{H}, \hat{L}^2] = 0, [\hat{H}, \hat{L}_z] = 0, \text{ 共同本征函数 } \psi_{nlm}(r, \theta, \varphi) (?)$$

$$\hat{H}, \hat{L}^2, \hat{L}_z \text{ 的本征值依次是 } E_n, l(l+1)\hbar, m\hbar$$

它的本征函数是 $\delta(x-x_0)$, 本征值是 x_0

力学量完全集 $\{\hat{A}_n | n \in \mathbb{N}, (\forall n \neq m, [\hat{A}_n, \hat{A}_m] = 0)\}$

共同完全集 ~~$\{\phi_{n_x n_y n_z}\}$~~ $\{\phi_{n_x n_y n_z}\}$, $\{n_x, n_y, n_z\}$ 为任一状态

$$\text{基态: } [\hat{x}, \hat{y}, \hat{z}], \quad \phi_{x_0 y_0 z_0} = \delta(x-x_0) \delta(y-y_0) \delta(z-z_0)$$

$$\text{动量: } \{\hat{p}_x, \hat{p}_y, \hat{p}_z\}, \quad \phi_{p_x p_y p_z} = (2\pi\hbar)^{-\frac{3}{2}} e^{\frac{i}{\hbar} \vec{p} \cdot \vec{r}}$$

坐标: $[\hat{L}^2, \hat{L}_z]$

$$\text{函数正交 } \int \phi_1^* \phi_2 dx = 0$$

Hermite 算符的本征函数:

$$\text{正交: } (\phi_{\lambda}, \phi_{\lambda'}) = \delta_{\lambda\lambda'}$$

$$\text{连续: } (\phi_{\lambda}, \phi_{\lambda'}) = \delta(\lambda - \lambda')$$

$$\text{完备性: 分立: } \hat{F}\phi_n(x) = a_n \phi_n(x)$$

$$\forall \psi, \quad \psi = \sum_n C_n \phi_n, \quad C_n = (\phi_n, \psi)$$

$$\psi(x) = \sum_n C_n \phi_n(x)$$

$$= \sum_n (\phi_n, \psi) \phi_n$$

$$= \sum_n \left(\int_{-\infty}^{+\infty} \phi_n^*(t) \psi(t) dt \right) \phi_n(x)$$

$$= \int_{-\infty}^{+\infty} \psi(t) \left(\sum_n \phi_n^*(t) \phi_n(x) \right) dt$$

$\downarrow$
 $\delta(t-x)$

$$\begin{aligned} & \int_{-\infty}^{+\infty} \psi(t) \left(\int_{-\infty}^{+\infty} \phi_{\lambda}^*(t) \phi_{\lambda}(x) dt \right) d\lambda \\ & \quad \downarrow \\ & \int_{-\infty}^{+\infty} \psi(t) \delta(t-x) dt = \psi(x) \end{aligned}$$

$$\text{连续: } \hat{F}\phi_{\lambda} = \lambda \phi_{\lambda}$$

$$\psi(x) = \int C(\lambda) \phi_{\lambda}(x) d\lambda \quad (\phi_{\lambda'}, \phi_{\lambda}) = \delta(\lambda' - \lambda)$$

$$C(\lambda) = \int \psi(x) \phi_{\lambda}^*(x) dx \Rightarrow \psi(x) = \int_{-\infty}^{+\infty} \left(\int_{-\infty}^{+\infty} \psi(t) \phi_{\lambda}^*(t) dt \right) \phi_{\lambda}(x) d\lambda$$

量子統計

2021.10.09.

力學量的測量

分立. 在 ψ_n 態中, $\bar{A} = (\psi_n, \hat{A} \psi_n) = A_n (\psi_n, \psi_n) = 1$, $\{\psi_n\}$ 正交歸一

$$\psi = \sum_n C_n \psi_n$$

$$\begin{aligned} \text{在 } \psi \text{ 態中, } \bar{A} &= (\psi, \hat{A} \psi) = \left(\sum_n C_n \psi_n, \sum_n C_n A_n \psi_n \right) \\ &= \sum_n |C_n|^2 A_n \end{aligned}$$

連續. $\psi(x) = \int C_\lambda \phi_\lambda(x) d\lambda$, $C_\lambda = \int \phi_\lambda^*(x) \psi(x) dx$

$$|C_\lambda|^2 = |(\phi_\lambda, \psi)|^2, \text{ 測量值在 } \lambda \sim \lambda + d\lambda \text{ 的概率密度}$$

多自由度

$$\hat{A} \phi_{\lambda n} = \lambda_n \phi_{\lambda n} \quad (\lambda = 1, 2, \dots, f_n)$$

$$1 \leftrightarrow 2\pi\delta(\omega)$$

$$\psi = \sum_n \sum_{m=1}^{f_n} C_{n,m} \phi_{nm}$$

$$\int_{-\infty}^{+\infty} \exp(-i\omega t) dt$$

$$\psi_{\lambda n} = \sum_{m=1}^{f_n} C_{n,m} \phi_{nm}$$

$$W(\lambda_n) = |(\psi_{\lambda n}, \hat{A} \psi)|^2 = \sum_{m=1}^{f_n} |C_{n,m}|^2$$

在定態上, 任何不顯含時間的力學量測量值分佈概率不隨時變化

平均值, $\bar{F} = \frac{(\psi, F\psi)}{(\psi, \psi)}$, 此時 ψ 不一定要歸一

動量算符

$$\hat{p} = -i\hbar \nabla, \text{ 本征方程 } -i\hbar \nabla \psi = \vec{p} \psi$$

$$-i\hbar \frac{\partial \psi}{\partial x_n} = p_{x_n} \psi, \text{ 利用分離變量法 } \psi = \prod_n \psi_n$$

$$\Rightarrow -i\hbar \frac{d\psi_n}{dx_n} = p_{x_n} \psi_n \Rightarrow \ln \psi_n = i \frac{p_{x_n}}{\hbar} x_n + C_n$$

$$\Rightarrow \psi_n = C_n \exp\left(\frac{i}{\hbar} p_{x_n} x_n\right)$$

$$\Rightarrow \psi = C \exp\left(\frac{i}{\hbar} \vec{p} \cdot \vec{r}\right), \quad \cancel{\psi = C \exp\left(\frac{i}{\hbar} \vec{p} \cdot \vec{r}\right)}$$

~~$\psi(\vec{r})$~~

$$\int_{\infty} \psi_{\vec{p}'}^*(\vec{r}) \psi_{\vec{p}}(\vec{r}) d\vec{r} = |C|^2 \prod_{k=1}^n \int_{-\infty}^{+\infty} \exp\left(i \frac{p_{x_k}' - p_{x_k}}{\hbar} x_k\right) dx_k$$

$$= |C|^2 \prod_{k=1}^n \hbar \int_{-\infty}^{+\infty} \exp(i(p_{x_k}' - p_{x_k}) x_k') dx_k'$$

$$= |C|^2 \prod_{k=1}^n 2\pi\hbar \delta(p_{x_k}' - p_{x_k})$$

$$= |C|^2 (2\pi\hbar)^n \delta(\vec{p}' - \vec{p}), \quad C = (2\pi\hbar)^{-\frac{n}{2}}$$

箱归一化

若粒子在束缚态, (在 $L \times L \times L$ 的空间中存在), 则需满足周期性边界条件 $[-\frac{L}{2}, \frac{L}{2}]^3$

$$\exp\left(\frac{i}{\hbar} p_{zn} L\right) = 1 \Leftrightarrow p_{zn} = \frac{2\pi m_{zn} \hbar}{L}, \quad m_{zn} \in \mathbb{Z}$$

角动量算符

$$\hat{L} = \vec{r} \times \vec{p} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ \hat{p}_x & \hat{p}_y & \hat{p}_z \end{vmatrix}$$

球坐标

$$\hat{L}_x = i\hbar \left(\sin\varphi \frac{\partial}{\partial\theta} + \cot\theta \cos\varphi \frac{\partial}{\partial\varphi} \right)$$

$$\hat{L}_y = -i\hbar \left(\cos\varphi \frac{\partial}{\partial\theta} - \cot\theta \sin\varphi \frac{\partial}{\partial\varphi} \right)$$

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial\varphi}$$

$$\hat{L}^2 = -\hbar^2 \left(\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial}{\partial\theta} \right) + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\varphi^2} \right)$$

$\hat{L}_z$ 的本征方程

$$\hat{L}_z \psi(\varphi) = L_z \psi(\varphi) = -i\hbar \frac{d\psi}{d\varphi} \Rightarrow \psi(\varphi) = C \exp\left(\frac{i}{\hbar} L_z \varphi\right)$$

$$\psi(2\pi) = \psi(0) \Rightarrow \frac{2\pi L_z}{\hbar} = 2\pi m_z \Rightarrow L_z = m_z \hbar$$

$$\int_0^{2\pi} |\psi|^2 d\varphi = 2\pi |C|^2 = 1 \Rightarrow C = \frac{1}{\sqrt{2\pi}}$$

$\hat{L}^2$ 的本征方程

$$\hat{L}^2 Y(\theta, \varphi) = \lambda \hbar^2 Y(\theta, \varphi), \quad Y(\theta, \varphi) = \Theta(\theta) \exp(i m \varphi)$$

$$-\left[\frac{1}{\sin\theta} \exp(i m \varphi) \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial \Theta}{\partial\theta} \right) + \frac{1}{\sin^2\theta} (-i m)^2 \Theta \exp(i m \varphi) \right] = \lambda \Theta \exp(i m \varphi)$$

$$\Rightarrow \frac{d}{d\xi} \left((1-\xi^2) \frac{d\Theta(\xi)}{d\xi} \right) + \left(\lambda - \frac{m^2}{1-\xi^2} \right) \Theta(\xi) = 0$$

$$\lambda = l(l+1), \quad l = |m|, |m|+1, \dots$$

$$\Theta(\xi) = P_l^m(\xi) = \frac{1}{2^l l!} (1-\xi^2)^{\frac{m}{2}} \frac{d^{l+m}}{d\xi^{l+m}} (\xi^2-1)^l \quad \text{Legendre 多项式}$$

$$l = 0, 1, 2, \dots \quad m = -l, -l+1, \dots, 0, \dots, l-1, l$$

$$\Theta(\xi) = N_{lm} P_l^m(\xi)$$

$$Y_{lm}(\theta, \varphi) = N_{lm} P_l^m(\cos\theta) \exp(i m \varphi)$$

$$\hat{L}^2 Y_{lm} = l(l+1) \hbar^2 Y_{lm}$$

$$\hat{L}_z Y_{lm} = m \hbar Y_{lm}$$

$$\int Y_{l'm'}(\theta, \varphi) Y_{lm}(\theta, \varphi) d\Omega = \delta_{l'l} \delta_{m'm} d\Omega = \sin\theta d\theta d\varphi$$

$l \rightarrow$ 軌道角量子數 $m \rightarrow$ 磁量子數

$$\text{定軸轉動 } H = \frac{L_z^2}{2J} \Rightarrow \hat{H} = \frac{\hat{L}_z^2}{2J}, \quad \psi(\varphi) = \exp(i m \varphi)$$

$$E_m = \frac{(m\hbar)^2}{2J}$$

$$\text{定點轉動 } H = \frac{L^2}{2J} \Rightarrow \hat{H} = \frac{\hat{L}^2}{2J}, \quad \psi(\theta, \varphi) = Y_{lm}(\theta, \varphi), \quad E_{lm} = \frac{l(l+1)\hbar^2}{2J}$$

$$l = 0, 1, 2, \dots$$

$$m = -l, -l+1, \dots, l-1, l$$

2021.10.13

守恒量: $\bar{A} = (\psi, \hat{A} \psi)$, $\frac{dA}{dt} = \left(\frac{\partial \psi}{\partial t}, \hat{A} \psi \right) + \left(\psi, \frac{\partial \hat{A}}{\partial t} \psi \right) + \left(\psi, \hat{A} \frac{\partial \psi}{\partial t} \right)$

$$= -\frac{1}{i\hbar} (\hat{H} \psi, \hat{A} \psi) + \frac{\partial \bar{A}}{\partial t} + \frac{1}{i\hbar} (\psi, \hat{A} \hat{H} \psi)$$

$$= \frac{\partial \bar{A}}{\partial t} + \frac{1}{i\hbar} [\hat{A}, \hat{H}]$$

若 $\frac{\partial \bar{A}}{\partial t} = 0$, 则 $\frac{dA}{dt} = \frac{1}{i\hbar} [\hat{A}, \hat{H}]$, $[\hat{A}, \hat{H}] = 0 \Rightarrow \frac{dA}{dt} = 0$

在任意态 $\psi(t)$ 下 A 的概率分布不变

在中心力场中,

$$[\hat{L}_z, \hat{H}] = \frac{1}{2m} [\hat{L}_z, \vec{p}^2] + [\hat{L}_z, V]$$

$$= \frac{1}{2m} [\hat{L}_z, \hat{p}_x^2] + \frac{1}{2m} [\hat{L}_z, \hat{p}_y^2] + [\hat{L}_z, V]$$

$$= \frac{1}{2m} (\hat{p}_x [\hat{L}_z, \hat{p}_x] + [\hat{L}_z, \hat{p}_x] \hat{p}_x) + \frac{1}{2m} (\hat{p}_y [\hat{L}_z, \hat{p}_y] + [\hat{L}_z, \hat{p}_y] \hat{p}_y) + [\hat{L}_z, V]$$

$$= \frac{1}{2m} [\hat{L}_z, \hat{p}_x \hat{p}_y + \hat{p}_y \hat{p}_x - \hat{p}_y \hat{p}_x - \hat{p}_x \hat{p}_y] + 0$$

$$= 0$$

$\psi(t) = \sum_k a_k(t) \psi_k$, $\hat{H} \psi_k = E_k \psi_k$, $\hat{A} \psi_k = A_k \psi_k$ (因为 $[\hat{A}, \hat{H}] = 0$, 所以有共同本征函数).
 $(\psi_k, \psi_m) = \delta_{k,m}$

$a_k(t) = (\psi_k, \psi(t))$, $|a_k(t)|^2$ 为测得 A_k 的概率

$$\frac{d|a_k(t)|^2}{dt} = \frac{da_k^*}{dt} a_k + \text{c.c.}$$

$$= \left(\frac{\partial \psi}{\partial t}, \psi_k \right) (\psi_k, \psi) + \text{c.c.}$$

$$= -\frac{1}{i\hbar} (\hat{H} \psi, \psi_k) (\psi_k, \psi) + \text{c.c.}$$

$$= -\frac{E_k}{i\hbar} |(\psi, \psi_k)|^2 + \text{c.c.} = 0.$$

不确定度: 偏差算符 $\Delta \hat{A} = \hat{A} - \bar{A}$, $\overline{\Delta \hat{A}} = \overline{(\hat{A} - \bar{A})} = \bar{A} - \bar{A} = 0$

$$\overline{\Delta \hat{A}^2} = \overline{(\hat{A} - \bar{A})^2} = \overline{\hat{A}^2 - 2\bar{A}\hat{A} + \bar{A}^2} = \overline{\hat{A}^2} - 2\bar{A}\bar{A} + \bar{A}^2 = \overline{\hat{A}^2} - \bar{A}^2$$

$$\Delta A = \sqrt{(\Delta A)^2} = \sqrt{\overline{\hat{A}^2} - \bar{A}^2}$$

不确定关系 $\Delta A \cdot \Delta B \geq \frac{1}{2} |\overline{[\hat{A}, \hat{B}]}| = \frac{1}{2} |(\psi, [\hat{A}, \hat{B}] \psi)|$

证明思路: 构造函数 $I(\lambda) = \int_{-\infty}^{+\infty} |\psi(\Delta \hat{A} + i\lambda \Delta \hat{B})|^2 dx \geq 0$

柯西不等式证明方法

量子统计

2.1.10.15

中心力场 $V(r)$ 具有转动对称性, $[L, H] = 0$

一般力学量完全集 $\{H, L^2, L_z\}$ 用 ψ_{nlm} 表示其本征态

$$H \psi_{nlm} = E_n \psi_{nlm}$$

$$L^2 \psi_{nlm} = L(L+1) \hbar^2 \psi_{nlm}$$

$$L_z \psi_{nlm} = m \hbar \psi_{nlm}$$

在球坐标中,

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \nabla_{\theta\varphi}^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \left(\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right)$$

$$\hat{H} = \frac{\hat{p}^2}{2\mu} = -\frac{\hbar^2}{2\mu} \nabla^2 = -\frac{\hbar^2}{2\mu} \left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} \right)$$

$$\nabla_{\theta\varphi}^2 = \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} = -\frac{\hat{L}^2}{\hbar^2}$$

$$\hat{H} = \frac{\hat{p}^2}{2\mu} + \frac{\hat{L}^2}{2\mu r^2} + V(r)$$

不妨设 $\psi_{nlm}(r, \theta, \varphi) = R(r) Y_{lm}(\theta, \varphi)$

$$\left[-\frac{1}{r} \frac{d^2}{dr^2} r + \frac{\mu}{\hbar^2} (E - V(r)) - \frac{l(l+1)}{r^2} \right] R(r) = 0$$

两体问题 设 $\frac{\partial}{\partial t} \psi(\vec{r}_1, \vec{r}_2, t) = \left[-\frac{\hbar^2}{2m_1} \nabla_{\vec{r}_1}^2 - \frac{\hbar^2}{2m_2} \nabla_{\vec{r}_2}^2 + V(|\vec{r}_1 - \vec{r}_2|) \right] \psi(\vec{r}_1, \vec{r}_2, t)$

质心 $\vec{r} = \vec{r}_1 - \vec{r}_2$, 质心坐标 $\vec{R} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{M}$

约化质量 $\mu = \frac{m_1 m_2}{m_1 + m_2}$

$(\vec{r}_1, \vec{r}_2) \leftrightarrow (\vec{r}, \vec{R})$

$$\frac{\partial}{\partial x_1} = \frac{\partial X}{\partial x_1} \frac{\partial}{\partial X} + \frac{\partial x}{\partial x_1} \frac{\partial}{\partial x}$$

$$= \frac{m_1}{M} \frac{\partial}{\partial X} + \frac{\partial}{\partial x}$$

$$\frac{\partial^2}{\partial x_1^2} = \frac{m_1^2}{M^2} \frac{\partial^2}{\partial X^2} + \frac{\partial^2}{\partial x^2} + \frac{2m_1}{M} \frac{\partial^2}{\partial X \partial x}$$

$$\nabla_{\vec{r}_1}^2 = \frac{m_1^2}{M^2} \nabla_{\vec{R}}^2 + \nabla_{\vec{r}}^2 + \frac{2m_1}{M} \nabla_{\vec{R}} \cdot \nabla_{\vec{r}}$$

$$\frac{\partial}{\partial x_2} = \frac{\partial X}{\partial x_2} \frac{\partial}{\partial X} + \frac{\partial x}{\partial x_2} \frac{\partial}{\partial x}$$

$$= \frac{m_2}{M} \frac{\partial}{\partial X} - \frac{\partial}{\partial x}$$

$$\frac{\partial^2}{\partial x_2^2} = \frac{m_2^2}{M^2} \frac{\partial^2}{\partial X^2} + \frac{\partial^2}{\partial x^2} - \frac{2m_2}{M} \frac{\partial^2}{\partial X \partial x}$$

$$\nabla_{\vec{r}_2}^2 = \frac{m_2^2}{M^2} \nabla_{\vec{R}}^2 + \nabla_{\vec{r}}^2 - \frac{2m_2}{M} \nabla_{\vec{R}} \cdot \nabla_{\vec{r}}$$

$$\hat{T} = -\frac{\hbar^2}{2} \left(\frac{1}{M} \nabla_{\vec{R}}^2 + \frac{1}{\mu} \nabla_{\vec{r}}^2 \right)$$

$$\hat{H}_{\vec{R}} = -\frac{\hbar^2}{2M} \nabla_{\vec{R}}^2, \quad \hat{H}_{\vec{r}} = -\frac{\hbar^2}{2\mu} \nabla_{\vec{r}}^2 + V(|\vec{r}|)$$

量子統計 [第三次作業]

2021. 10. 17

廖收鋒 級91 集電路學院 2019010896

$$2. (3.1) \quad \psi(x) = \sqrt{\frac{\alpha}{\pi}} \exp\left(-\frac{\alpha^2 x^2}{2} - \frac{i}{2} \omega t\right), \quad \alpha = \sqrt{\frac{m\omega}{\hbar}}, \quad (\psi, \psi) = 1$$

$$\begin{aligned} (1) \quad \bar{U} &= (\psi, \hat{U} \psi) = (\psi, U \psi) = \int_{-\infty}^{+\infty} \frac{1}{2} m \omega^2 x^2 \cdot \frac{\alpha}{\pi} \exp(-\alpha^2 x^2) dx \\ &= \int_{-\infty}^{+\infty} \frac{1}{2} \cdot \frac{m\omega^2}{\pi \alpha^2} \xi^2 \exp(-\xi^2) d\xi \\ &= \frac{\hbar\omega}{2\sqrt{\pi}} \int_{-\infty}^{+\infty} \xi^2 \exp(-\xi^2) d\xi \\ &= \frac{\hbar\omega}{4\sqrt{\pi}} \int_{-\infty}^{+\infty} \xi^2 \exp\left(-\frac{\xi^2}{2}\right) d\left(\frac{\xi^2}{2}\right) \\ &= \frac{\hbar\omega}{4\sqrt{\pi}} \left[\xi^2 \exp(-\xi^2) \Big|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} \exp(-\xi^2) d\xi \right] \\ &= \frac{\hbar\omega}{4\sqrt{\pi}} \cdot \int_{-\infty}^{+\infty} \exp(-\xi^2) d\xi \\ &= \frac{\hbar\omega}{4} \end{aligned}$$

$$\begin{aligned} (2) \quad \bar{T} &= (\psi, \hat{T} \psi) = \frac{1}{2m} (\psi, \hat{p}^2 \psi) = \frac{1}{2m} (\psi, -\hbar^2 \frac{d^2 \psi}{dx^2}) \\ &= -\frac{\hbar^2}{2m} (\psi, \frac{d^2 \psi}{dx^2}) = -\frac{\hbar^2}{2m} (\psi, -x^2 \frac{d}{dx} (x \psi)) \\ &= \frac{\hbar^2}{2m} (\psi, x^2 \frac{d \psi}{dx}) = \frac{\hbar^2 x^2}{2m} (\psi, \psi + x \frac{d \psi}{dx}) \\ &= \frac{\hbar^2}{2m} (\psi, \psi - \alpha^2 x^2 \psi) \\ &= \frac{\hbar^2 x^2}{2m} [1 - \alpha^2 (\psi, x^2 \psi)] \\ &= \frac{\hbar^2 x^2}{2m} \left[1 - \alpha^2 \cdot \frac{\bar{U}}{\frac{1}{2} m \omega^2} \right] = \frac{1}{2} \cdot \hbar\omega \left[1 - 2 \cdot \frac{\hbar\omega}{\hbar} \cdot \frac{\hbar\omega}{4\hbar\omega^2} \right] = \frac{\hbar\omega}{4} \end{aligned}$$

3. (3.2)

~~(\psi, \psi) =~~

$$|\psi(r)|^2 = |\psi(r, \theta, \varphi)|^2 \cdot 4\pi r^2 = \frac{1}{\pi a_0^3} \exp\left(-\frac{2r}{a_0}\right) \cdot 4\pi r^2 = \frac{4}{a_0^3} \exp\left(-\frac{2r}{a_0}\right) r^2$$

$$(\psi, \psi) = \int_0^\infty \frac{1}{\pi a_0^3} \exp\left(-\frac{2r}{a_0}\right) \cdot 4\pi r^2 dr = \frac{1}{2} \int_0^\infty \exp(-\xi) \xi^2 d\xi$$

$$= \frac{1}{2} \int_0^\infty 2\xi \exp(-\xi) d\xi = \int_0^\infty \xi \exp(-\xi) d\xi = \int_0^\infty \exp(-\xi) d\xi = 1$$

$$(1) \bar{r} = \int_0^\infty r |\psi(r)|^2 dr = \int_0^\infty \frac{4}{a_0^3} \exp\left(-\frac{2r}{a_0}\right) \cdot \left(\frac{r}{a_0}\right)^2 dr = \frac{a_0}{4} \int_0^\infty \exp(-\xi) \xi^3 d\xi$$

$$= \frac{a_0}{4} \int_0^\infty \xi^2 \exp(-\xi) d\xi = \frac{3a_0}{4} \cdot \frac{1}{2} \int_0^\infty \exp(-\xi) \xi^2 d\xi = \frac{3}{2} a_0$$

$$(2) \overline{\frac{e^2}{r}} = -e^2 \int_0^\infty \frac{4}{a_0^3} \exp\left(-\frac{2r}{a_0}\right) r dr = -e^2 \cdot \frac{1}{a_0} \int_0^\infty \exp(-\xi) \xi d\xi$$

$$= -\frac{e^2}{a_0}$$

$$(3) \frac{d|\psi(r)|^2}{dr} = \frac{4}{a_0^3} \left[2r \exp\left(-\frac{2r}{a_0}\right) - \frac{2}{a_0} r^2 \exp\left(-\frac{2r}{a_0}\right) \right]$$

$$= \frac{8r \exp\left(-\frac{2r}{a_0}\right)}{a_0^3} \left(1 - \frac{r}{a_0} \right), \quad \frac{d|\psi(r)|^2}{dr} \Big|_{r=r_0} = 0$$

$\Rightarrow$ 最可几半径 $r_0 = a_0$

$$(1) \bar{T} = (\psi, \hat{T} \psi) = \frac{\hbar^2}{2m} (\psi, \frac{d}{dr} \frac{d\psi}{dr}) = \frac{\hbar^2}{2ma_0} (\psi, \psi) = \frac{\hbar^2}{2ma_0}$$

$$\hat{T} = -\frac{\hbar^2}{2m} \nabla^2 + \frac{\hbar^2}{2m}, \quad \hat{L} \psi = 0$$

$$\nabla^2 = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2}, \quad \frac{\partial \psi}{\partial \theta} = 0$$

$$\bar{T} = (\psi, \hat{T} \psi) = (\psi, -\frac{\hbar^2}{2mr} \frac{\partial}{\partial r} (r \frac{\partial \psi}{\partial r}))$$

$$= (\psi, -\frac{\hbar^2}{2mr} \frac{\partial}{\partial r} (-\frac{r}{a_0} \psi)) = \frac{\hbar^2}{2ma_0} (\psi, \frac{1}{r} \frac{\partial}{\partial r} (r \psi))$$

$$= \frac{\hbar^2}{2ma_0} (\psi, \frac{\psi}{r} + \frac{\partial \psi}{\partial r})$$

$$= \frac{\hbar^2}{2ma_0} \left(\frac{1}{r} - \frac{1}{a_0} \right)$$

$$(1) \hat{T} = -\frac{\hbar^2}{2m} \nabla^2 + \frac{\hbar^2}{2I}, \quad \hat{L} \psi = 0, \quad \nabla^2 = \frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \hat{f}(\theta, \varphi), \quad \hat{f}(\theta, \varphi) \psi = 0$$

$$\Rightarrow \bar{T} = (\psi, \hat{T} \psi) = -\frac{\hbar^2}{2m} (\psi, (\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r}) \psi)$$

$$= -\frac{\hbar^2}{2m} (\psi, (\frac{1}{a_0^2} - \frac{2}{a_0 r}) \psi) = -\frac{\hbar^2}{2m} \cdot \frac{1}{a_0^2} - \frac{1}{a_0 r}$$

$$= -\frac{\hbar^2}{2m} \left(\frac{1}{a_0^2} - \frac{2}{a_0^2} \right) = \frac{\hbar^2}{2ma_0^2}$$

4. (3.5) (1) $\hat{H} = \frac{\hat{L}_z^2}{2I}$

$$\hat{H}\psi = E\psi = \frac{\hat{L}_z^2}{2I}\psi = \frac{-\hbar^2}{2I} \frac{d^2}{d\varphi^2}\psi$$

$$\psi = A \exp(i \cdot \frac{\sqrt{2IE}}{\hbar} \varphi) + B \exp(-i \cdot \frac{\sqrt{2IE}}{\hbar} \varphi)$$

$$\text{令 } \psi(\varphi+2\pi) = \psi(\varphi), \quad \psi'(\varphi+2\pi) = \psi'(\varphi)$$

$$A+B = A \exp(i \cdot 2\pi \cdot \frac{\sqrt{2IE}}{\hbar}) + B(-i \cdot 2\pi \cdot \frac{\sqrt{2IE}}{\hbar})$$

$$i \frac{\sqrt{2IE}}{\hbar^2} (A-B) =$$

因为 $[\hat{L}_z, \hat{L}_z^2] = 0$, 所以两者具有共同本征函数。

$\hat{L}_z$ 的本征函数 $\psi_m = \frac{1}{\sqrt{2\pi}} e^{im\varphi}$, 本征值 $L_{z,m} = m\hbar$, $m \in \mathbb{N}$.

$$\hat{H}\psi_m = \frac{\hat{L}_z^2}{2I}\psi_m = \frac{\hbar^2}{2I} \psi_m = \frac{\hat{L}_z}{2I} (\hat{L}_z \psi) = \frac{m^2 \hbar^2}{2I} \psi_m$$

$$\text{定态能量为 } E_m = \frac{m^2 \hbar^2}{2I}$$

$$\text{定态波函数为 } \psi_m = \frac{1}{\sqrt{2\pi}} e^{im\varphi}$$

(2) $\hat{H} = \frac{\hat{L}^2}{2I}$, 因为 $\hat{L}$ 的本征波函数是球谐函数 $Y_{lm}(\theta, \varphi)$

$$\text{本征值是 } l(l+1)\hbar^2, \quad m = -l, -l+1, \dots, l-1, l$$

$$\text{所以 } \hat{H} \text{ 的本征值 (定态能量) 是 } \frac{l(l+1)\hbar^2}{2I}$$

$$\text{本征波函数是 } Y_{lm}(\theta, \varphi)$$

$$5. (3.1) \quad \psi(x) = A \left[\sin^2 kx + \frac{1}{2} \cos kx \right] = \frac{A}{2} [1 + \cos kx - \cos 2kx] = \frac{A}{2} \left[-\frac{1}{2} e^{-i \cdot 2kx} + \frac{1}{2} e^{-i kx} + 1 + \frac{1}{2} e^{i kx} - \frac{1}{2} e^{i \cdot 2kx} \right]$$

$$= \frac{A}{4} \sqrt{2\pi\hbar} \left(\frac{1}{\sqrt{2\pi\hbar}} e^{i \cdot (-2k)x} - \frac{1}{\sqrt{2\pi\hbar}} e^{i(-k)x} + \frac{2}{\sqrt{2\pi\hbar}} e^{i \cdot 0x} + \frac{1}{\sqrt{2\pi\hbar}} e^{i kx} - \frac{1}{\sqrt{2\pi\hbar}} e^{i \cdot 2kx} \right)$$

$$k = \frac{p}{\hbar}$$

共有5個態，可測得的概率依次為 $\frac{1}{8}, \frac{1}{8}, \frac{1}{2}, \frac{1}{8}, \frac{1}{8}$

$$\bar{p} = (-2\hbar k) \times \frac{1}{8} + (-\hbar k) \times \frac{1}{8} + 0 \times \frac{1}{2} + (\hbar k) \times \frac{1}{8} + (2\hbar k) \times \frac{1}{8} = 0$$

$$\bar{T} = \frac{1}{8} \frac{1}{2m} \left[(2\hbar k)^2 \times \frac{1}{8} \times 2 + (\hbar k)^2 \times \frac{1}{8} \times 2 + 0^2 \times \frac{1}{2} \right] = \frac{5\hbar^2 k^2}{8m}$$

$$6. (3.1) \quad \psi(x) = Ax(a-x) = C \sum_{n=1}^{\infty} \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right)$$

$$C a_n = \int_0^a Ax(a-x) \cdot \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right) dx$$

$$= A a^3 \int_0^1 \frac{x}{a} (1 - \frac{x}{a}) \sqrt{\frac{2}{a}} \left(\frac{n\pi}{a}\right) d\left(\frac{x}{a}\right)$$

$$= A a^3 \int_0^1 \xi (1 - \xi) \sqrt{\frac{2}{a}} \sin(n\pi \xi) d\xi$$

$$= A a^3 \sqrt{\frac{2}{a}} \int_0^1 \xi (1 - \xi) \sin(n\pi \xi) d\xi$$

$$= A a^3 \sqrt{\frac{2}{a}} \cdot \frac{1}{n\pi} \left[\xi (1 - \xi) \cos(n\pi \xi) \Big|_0^1 + \int_0^1 \cos(n\pi \xi) (1 - 2\xi) d\xi \right]$$

$$= A a^3 \sqrt{\frac{2}{a}} \cdot \left(\frac{1}{n\pi}\right)^2 \left[(1 - 2\xi) \sin(n\pi \xi) \Big|_0^1 + 2 \int_0^1 \sin(n\pi \xi) d\xi \right]$$

$$= 2 A a^3 \sqrt{\frac{2}{a}} \left(\frac{1}{n\pi}\right)^3 [1 - (-1)^n]$$

$$\psi(x) = \sum_{n=1}^{\infty} 4 A a^3 \sqrt{\frac{2}{a}} \left(\frac{1}{(2n-1)\pi}\right)^3 \sin\left(\frac{(2n-1)\pi}{a}x\right)$$

$$1 = \int_0^a |\psi(x)|^2 dx = \int_0^a |A|^2 (x^2 - 2ax + a^2) dx = |A|^2 \left(\frac{1}{3} a^3 - 2a \cdot \frac{1}{4} a^4 + a^2 \cdot \frac{1}{3} a^3 \right)$$

$$= |A|^2 \left(\frac{1}{30} a^5 \right)$$

$$\Rightarrow |A| = \sqrt{\frac{30}{a^5}}$$

$$\psi(x) = \sum_{n=1}^{\infty} \frac{\sqrt{960}}{[(2n-1)\pi]^3} \sqrt{\frac{2}{a}} \sin\left(\frac{(2n-1)\pi}{a}x\right)$$

$$W(E_n) = \begin{cases} \left[\frac{\sqrt{960}}{(n\pi)^3} \right]^2 = \frac{960}{n^6 \pi^6}, & 2|n \\ 0, & 2 \nmid n \end{cases} \quad \left[\sum_{n=1}^{\infty} \frac{1}{(2n-1)^6} = \frac{\pi^6}{960} \right]$$

$$\begin{aligned}\bar{E} &= \int_0^a \psi^* \hat{H} \psi dx = \int_0^a \psi^* \cdot \left(-\frac{\hbar^2}{2m}\right) \frac{d^2\psi}{dx^2} dx = \cancel{\int_0^a \psi^* \cdot \left(-\frac{\hbar^2}{2m}\right) \frac{d^2\psi}{dx^2} dx} - \frac{\hbar^2}{2m} |A|^2 \int_0^a x(a-x) \cdot (-2) dx \\ &= \frac{\hbar^2}{m} |A|^2 \int_0^a (ax - x^2) dx \\ &= \frac{\hbar^2}{m} |A|^2 \left(\frac{1}{2} a^2 - \frac{1}{3} a^2 \right) = \frac{1}{6} \cdot \frac{\hbar^2 a^2}{m} \cdot \frac{30}{a^3} = \frac{5\hbar^2 a}{6ma^2}\end{aligned}$$

$$\left[\sum_{n=1}^{\infty} \frac{(2n-1)^2 \hbar^2 \pi^2}{2mc a^2} \cdot \frac{9\hbar^2}{(2n-1)^6 \hbar^6} = \frac{5\hbar^2}{ma^2} \Leftrightarrow \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} = \frac{\pi^4}{96} \right]$$

7. (3.9) 氢原子能量: $E_2 = -\frac{m_e e^2 \psi}{8\hbar^2} = \bar{E}$, 概率为 1

角动量平方: $L^2 = \ell(\ell+1)\hbar^2 = 2\hbar^2 = L^2$, 概率为 1

轴向角动量: $L_z = 0$, 概率为 $\frac{1}{4}$ 期望值 $\bar{L}_z = -\frac{3}{4}\hbar$
 $L_z = -\hbar$, 概率为 $\frac{3}{4}$.

8. $\hat{H} = \frac{1}{2I_1} (\hat{L}_x^2 + \hat{L}_y^2 + \frac{1}{2} \hat{L}_z^2) + \frac{1}{2I_2} \hat{L}_z^2 = \frac{\hat{L}^2}{2I_1} + \left(\frac{1}{2I_2} - \frac{1}{4I_1} \right) \hat{L}_z^2$

其本征函数是 $Y_{\ell m}(\theta, \varphi)$

$$\hat{H} Y_{\ell m}(\theta, \varphi) = \left[\frac{1}{2I_1} \ell(\ell+1)\hbar^2 + \left(\frac{1}{2I_2} - \frac{1}{4I_1} \right) m^2 \hbar^2 \right] Y_{\ell m}(\theta, \varphi)$$

对应本征值是 $E_{\ell m} = \left[\frac{1}{2I_1} \ell(\ell+1) + \left(\frac{1}{2I_2} - \frac{1}{4I_1} \right) m^2 \right] \hbar^2$

9. $[\hat{L}_z, \hat{L}_{\pm}] = [\hat{L}_z, \hat{L}_x \pm i\hat{L}_y] = [\hat{L}_z, \hat{L}_x] \pm i[\hat{L}_z, \hat{L}_y]$
 $= i\hbar \hat{L}_y \pm i(-i\hbar \hat{L}_x)$
 $= i\hbar \hat{L}_y \pm \hbar \hat{L}_x$
 $= \pm \hbar \hat{L}_x + i\hbar \hat{L}_y$
 $= \pm \hbar (\hat{L}_x \pm i\hat{L}_y) = \pm \hbar \hat{L}_{\pm}$

$$10. \bar{L}_x = (\psi_z, \hat{L}_x \psi_z)$$

$$= \frac{1}{i\hbar} (\psi_z, [\hat{L}_y, \hat{L}_z] \psi_z)$$

$$= \frac{1}{i\hbar} [(\psi_z, \hat{L}_y \hat{L}_z \psi_z) - (\psi_z, \hat{L}_z \hat{L}_y \psi_z)]$$

$$= \frac{1}{i\hbar} [(\hat{L}_y \psi_z, \hat{L}_z \psi_z) - (\hat{L}_z \psi_z, \hat{L}_y \psi_z)]$$

$$= \frac{1}{i\hbar} [(\hat{L}_y \psi_z, \hat{L}_z \psi_z) - (\hat{L}_y \psi_z, \hat{L}_z \psi_z)^*]$$

$$= \frac{L_z}{i\hbar} [(\hat{L}_y \psi_z, \psi_z) - (\hat{L}_y \psi_z, \psi_z)^*] = \frac{L_z}{i\hbar} (\overline{L_y}^* - \overline{L_y}) = 0$$

同理可得 $\overline{L_y} = 0$